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Proposing a novel heuristic algorithm for university course timetabling problem with the quality of courses rendered approach; a case study



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Abstract University course timetabling problem has been and is being considered as an operational issue by researchers. The purpose of University Course Timetabling Problem is to allocate academic courses to the lecturers, time slots and classrooms which must be executed in such a manner to promote to quality of education. In this context, a three-stage heuristic algorithm consisting of lecturers, time slots and classes is proposed. By applying the Days of lecturers' attendance to total days of planning ratio, Lecturer quality level in planning context, Offering the simultaneous prerequisite and main courses to the all course ratio, Courses presented in the last time-slot to total courses ratio, Surplus classroom capacity to classroom capacity ratio, Full-time lecturer to total lecturers ratio and Number of classrooms assigned to each lecturer to total available classroom ratio indexes, the planning obtained through this newly proposed algorithm is assessed and the response quality is determined. The available data of the subject university's industrial engineering department is fed in MATLAB software through this algorithm with the objective to have the University Course Timetabling Problem as the output. At its experimental stage, the output is measured through the same indexes revealing that course presentation by the lecturers is 96% of the possible optimum.

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1. Introduction

The subject of timetabling has been and is a major focus among researchers beginning 1980s [1]. The timetabling problem consists of different aspects like: transportation, sports, nurse timetabling and university timetabling [2]. Among the

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studies run, the focus of many is own University Course Timetabling Problem (UCTP), which is result through different proposed methods [3].

As to UCTP the objective can be optimized cost, transportation, quality etc. In many UCTPs some diverse objectives could be of concern, which can be formulated through multi criterion combinatorial optimization [4]. The UCTP can simply be defined as a series of occurrences (courses) assignable to time slots, where, some constraints unique to education system should be of concern, which in turn cause complexities in solving the problem [1].

The UCTP is a combinatorial optimization problem of NP-hard class, consisting of allocating courses, lecturers and students to a number of specific time slots and classrooms [5]. In general, the two sets of hard and soft constraints exist in this problem, where, the former must be fully satisfied at all conditions and the latter proportionally [1].

In UCTP, the first problem is in its being a NP-Complete problem and due to its exponential conditions, it cannot be solved through polynomial time classes, hence necessitating adoption of heuristic methods in solving it. The second problem here in its being complex due to the existent of different constraints unique to any academic institution, which must be of major concern. With this two issues at hand attempt is made by many researchers to propose different meta-heuristic and heuristic algorithms, where, the hard and soft constraints are of concern, to solve this problem [6,7].

In this study in addition to assessing the hard and soft constraints assessed in available literature, attempt is made to propose some new constraints within algorithm which will be fully explained in this article. The distinguishing features of this study consist of prioritization of full-time lecturer to part-time and lecturer performance quality.

This article is organized as follow: the research background is presented in Section 2; the heuristic algorithm is proposed in Section 3; the UCTP indexes assessment is run in Section 4; this proposed algorithm is implemented in Section 5 and the article is discussed and concluded in Section 6.

2. Research background

There exist many studies on UCTP's categories, where, in some the linear and nonlinear mathematical models and in others the meta-heuristic and heuristic methods are adopted.

Cacchiani et al. [8] presented a new method to solve UCTP and provided the planning in accordance with the student curriculum and that of the lecturers or lecturers. The constraints of concern there consist of provision of all curricula, lack of intrusiveness in the common courses, abstaining from intermingling in the planning in relation to the lecturers, classes and their capacities, crash courses and the minimizing the exchange of lecturers and students in the university.

Lewis and Thompson [9] proposed a mathematical behavior of UCTP due to its being NP-Hard, based on a two-stage algorithm where, the response space of the problem is of concern. Their findings indicate that for this purpose the neighborhood space, would improve. This algorithm illustrated that the quality of the obtained response has improved.

Soria-Alcaraz et al. [10] presented a locally based algorithm, in accordance with an iterative search, where, the UCTP is subject to two general conditions. In their assessments the two sets of hard and soft constraints are of concern. In the hard constraints the following prevail: the non-intermingling of classrooms, lack of intrusion in student planning and the lecturers' programming, and the absence of rendering different groups a course or lesson simultaneously and concurrently. In the soft constraints, issues like, lack of rendering courses within the last time-slot/day, an absence of scheduling more than 3 courses (lessons)/day for students, omission of individual or single courses/day for the students, classroom capacity, compactness (of schedule) on the days when the class is run and the presentation of the entire parts of every course or lesson in a session.

Mendez Diaz et al. [11] developed an UCTP by focusing on the details, where, a linear numerical planning model and an exploratory algorithm are involved. Their model and algorithm are implemented in a private university in Buenos Aires, Argentina and results indicate the execution capabilities and provide logical and qualitative responses.

As to solving the UCTP, Borchani et al. [12] presented an algorithm based on variable neighborhood descent search algorithms, with the objective to reduce the number of time intervals in UCTP time-slots and to reduce individual courses in the time-table of each group of students. In their algorithm, 11 distinctive neighborhood structures are of concern, 6 of which are applied to rectify the time-slot intervals and 5 for regulating the lesson or course period selected in the time-table. Their algorithm is implemented in a university of Tunisia and its capabilities is assessed with an outcome of 50% reduction in time intervals, which is a better performance of the university time-table.

Skoullis et al. [13] presented a Cat (swarm) Combinatorial Optimization algorithm, to solve the UCTP, where, the traditional CSO algorithm is applied after being modified. In their study, by applying data of Grecian high schools, their algorithm is implemented in the same. The result obtained from this algorithm are compared with that of PSO genetic algorithms, evolutionary, simulated annealing and artificial fish swarm algorithms. The modification of this algorithm makes CSO algorithms innovative, because it can be applied when a combination of responses face change.

As to UCTP, Song et al. [14] proposed an algorithm with an approach to optimize energy consumption efficiency, through which a 4% decrease is recorded in the cooling and heating energy consumption of the given university.

Bagger et al. [15] introduced a meta-heuristic method in preparing UCTP, where, UCTP is divided into two sub-problems: time-planning and classroom assignment, consequently, the Bender algorithm is applied. Their algorithm is based on the minimal cost. To verify the implementation of this method, 32 samples of related literature data are assessed and the results are compared with the results of the applied studies at large scale, indicating this method's outperformance.

According to Akkan and Gulcu [16], the traditional solutions in UCTP as to qualitative responses is not flexible, when subject to change as inputs. For this purpose they proposed a meta-heuristic multi-criteria algorithm. By applying a multi-

objective genetic algorithm, where the hill climbing and simulated annealing algorithms are involved in solving the problem. This algorithm is based on Pareto selection method, which yields quality responses.

In this context, a simulated annealing multi-objective algorithm is introduced by Gulcu and Akkan [17], through which, two cost function criterion and a robust function are defined. The parameters of the algorithm are in accordance with the iterated F-Race. This algorithm is tested in ITC-2007 educational planning, the results of which are compared with the multi-objective GA results and revealed its outperformance.

Myers and Yellen [18] developed a multi-objective UCTP, with the objective to first, refrain from lecturers planning intrusion and next to present a crash-course or a compact program for the students and the faculty. In their research, one certainty and two approximate algorithms are applied. This algorithm is implemented in Rollins University and revealed that this system is able to provide a compact UCTP.

Regarding a music school, Asgeirsson and Gunnarsdottir [19] run study with respect to UCTP. The unique feature of this school which has general and private courses is in its combining both high school and university planning. Here a numerical planning model is presented, the result of which indicate that, though, the timetabling proves to be challenging and nevertheless, by applying a two-stage approach, a highly qualitative time-table is obtained, in a sense that, priority is given to general session in the music school follow by private session.

In his study as to UCTP, Ramalho [20] presented a multi-purpose optimization solution, where, the game theory is applied to define the research constraint. This solution was implemented in Porto Engineering University in Portugal and the results indicate that when objectives and constraints contradict, as to the results the game theory becomes contributive.

Due to the vastness of the UCTP issue some of the existing relevant sources with the focus on method and constraints reviewed in this study are tabulated in Table 1.

In many of the available studies, where, a case-study is run and the problem is of a big size, with respect to the fact that, the problem is converted to a NP-hard, non-exact and approximate methods are applied, with a focus on meta-heuristic algorithms. In parallel with this issue, in this study, non-exact methods is applied to solve the model in the NP-hard mode. Due to the fact that, heuristic algorithms, contrary to meta-heuristic ones do not follow definite and general methods, are designed for UCTP in specific are of flexibility in solving this problem and at more iteration can secure a better response. Consequently, in this study, a heuristic algorithm is proposed as solution in solving the UCTP.

3. A heuristic algorithm as a solution for solving UCTP with a quality of offered courses approach

Though, heuristic algorithms, in contrast to meta-heuristic ones, are not able to solve all problem in general and are designed in accordance with the particular conditions of a problem are of better flexibility in accessing responses. The objective here is to determine an appropriate solution for UCTP at large scale, by proposing heuristic algorithm. Because, one of the most important indexes in assessing educa-

tional planning is education quality, this index is of major concern in this algorithm.

3.1. Hard and soft constraints in UCTP

As observed, in this context every research group or single researcher are concern with some constraint with respect to chosen methodology, model and specific implementation conditions. According to the concerned constraints in the available studies and applying an orientated-qualitative approach in this study, the hard and soft constraints are of concern here.

Hard Constraints consist of:

- *Scheduling for lecturers according to their hours of attendance:* the detail of tutorials for them only for the hours that they are present at the university
- *Lake of interference in lecturer scheduling:* a lecturer cannot tutor more than one course during a time-slot
- *Maximum permissible courses for each lecturer/day:* it is not feasible for any lecturer to tutor more than the specified hours/day
- *Presenting all curriculums on all subjects:* all the groups predicted in the program of each subject must be scheduled in the time-table
- *Lake of interference in allocating classrooms:* not more than one course should be assigned to one classroom in a time-slot.
- *Capacity of classroom:* courses are assigned to classrooms which have equal or more capacity for the student courses capacity
- *Considering the total tutoring hours of lecturers:* lecturers are assigned to specific hours for tutorials throughout the week

Soft constraints:

- *Prioritization of full-time lecturer to part-time:* lecturers of each university impose less educational costs for the university, than the part-time faculty staff and are more within access of the students; consequently in this model, priority is given to lecturers rather than part-time lecturer

Note 1: Lecturers are those who are employed on full-time bases as permanent lecturers and part-time lecturers are employed as temporary lecturer.

- *Lecturers' quality of teaching:* because one subject can be taught by many lecturers, the priority must be given to the one with higher quality
- *Utilizing the lecturer presence time at its maximum:* to reduce the time to and from university spent by the lecturer, in this model, lecturer presence time is utilized at its maximum

3.2. A heuristic algorithm for UCTP

In this context, a three-stage heuristic algorithm consisting of lecturers, time slots and classes is proposed.

Phase 1: Assigning of lecturers to courses

In the initial stage of this heuristic algorithm, the related lecturers to the courses are assigned with respect to increasing

Table 1 Research background summary.

Name of Author and Year	Method		Constraints										
	Exact	Approximate	Presenting entire curriculums of an individual course/lesson	Planning standard curriculums based on the hours of attendance	Refrain from intermingling in the program of lecturers	Considering the experience required for every course	Refraining from intermingling with respect to the location of classrooms	Classroom Capacity	Lack of interference in the students' program	Lack of concurrent classes for a group for one course	Considering standard capacities from the viewpoint of teaching	Running classes through the week in a compact manner	Lack of presenting lessons at the last time-slot/day
Al-Yakoob and Sherali [21]	X				✓								
Vermuyten et al. [22]	X				✓		✓		✓				
Soria-Alcaraz et al. [10]	X				✓		✓		✓	✓		✓	✓
Bellio et al. [23]		X			✓		✓	✓				✓	
Fonseca et al. [24]		X		✓	✓	✓	✓						
Saviniec and Constantino [25]		X	✓	✓	✓		✓						
Skoullis et al. [13]		X	✓		✓		✓					✓	✓
Akkan and Gulcu [16]		X	✓	✓	✓		✓						
Bagger et al. [15]		X	✓	✓	✓		✓	✓		✓			
Asgeirsson and Gunnarsdottir [19]	X		✓	✓	✓	✓			✓				
Gulcu and Akkan [17]		X	✓	✓	✓								
Myers and Yellen [18]		X	✓	✓	✓				✓			✓	

educational quality approach following specific order in the following steps:

Step 1: The courses of concern in this planning must be determined and prioritized with respect to the numbers of lecturers with the efficiency to run the courses. It is for this purpose that, the courses that fewer lecturer can teach are prioritized than other courses.

Step 2: Courses with higher priority are assigned to the lecturers first.

To begin with, the course within higher priority are selected and if there is only one lecturer with this ability, he is assigned to the course, otherwise go to the step 3.

Step 3: Lecturers who have proclaimed their hours of attendance at the university are prioritized according to the total teaching hours. If this priority is uniform among them, go to the step 5 otherwise, go to the step 4.

Step 4: The designated hours to the lecturers for the assigned courses in step 2 should be deducted from the total lecturing hours and they be prioritized for programming based on their hours of attendance. The process of this prioritization is as follows:

1. Divide the lecturers into two sets as full-time and part-time lecturers and determine the hours of their attendance in the university for each of these individuals.
2. Prioritize full-time lecturers based on their hours of attendance, which would determine: the lower the hours of their attendance, the higher their priority within 1 to n.
3. Prioritize part-time lecturers based on their hours of attendance, which would determine: the lower the hours of their attendance, the higher their priority within n + 1 to p.

Step 5: Assign lecturers to courses with higher priority (based on Step 1), for this purpose the following phases are of concern:

- Course i with a higher priority is selected.
- Lecturers who have the ability to teach this course are determined.
- In case, there is at least one full-time lecturer for course i with tutoring capabilities, this course will not be assigned to a part-time lecturer.
- In case the determined lecturers have met the minimum teaching hours, the one with higher quality and teaching capacity will be assigned to the course.
- In case the determined lecturers have not met the minimum teaching hours, the same will be assigned to the course.
- In case more than one of the appointed lecturers fails to meet the his minimum teaching hours, the following equation is calculated for course i and lecturer p :

$$\sigma_{ip} = (\text{minimum permissible hours for lecturer } p - \text{hours assigned for courses}) * Q_{ip}$$

σ_{ip} is obtained for every lecturer who is capable of teaching the given course i . The higher σ_{ip} for any lecturer, his assignment to course i .

The same process holds true for selecting next course priority.

Note 2: All required course groups must be assigned.

Note 3: The hours assigned for courses relevant to the part-time lecturer, should not exceed the hours stipulated by the university.

Phase 2: Time-slot assignment for Courses–Lecturers

The steps assigned to the time-slots in relation to courses–lecturers are as follows:

Step 6: Here, the three indexes of number of days of attendance by the lecturers, diversity of their courses and the total hours assigned to the courses are assigned to determine lecturer priority and the formulation of timetabling is made by applying priority of lecturers. To prioritize the lecturers, the following equation is applied:

$$\beta_p = \text{number of days of attendance by the lecturer } p$$

*diversity of courses of the lecturer p

*Total number of hours for courses assigned to lecturers p

The lower β_p , the higher lecturer priority. The time slots is developed base on lecturer prioritization, followed by assigning time slot to course base on lecturer presence, making all lecturers to be placed in the timetable according to their priority.

Note 4: There should not be any interference in the lecturer's time slots.

Phase 3: Assigning a classroom to a course–lecturer in each time-slot

In the third phase of the heuristic algorithm as to the course-lecturer aspect, each time-slot is determined in the first and second phases is assigned to the classroom allocation, through the following steps:

Step 7: At this step has to do with classroom allocation with the beginning on the first of the week. The courses are prioritized according to their capacity, in sense that the one with higher capacity has higher priority.

The following equation is applied to calculate the courses with higher priority:

$$\gamma_{im} = (\text{Capacity of classroom } m - \text{Capacity of Course } i)$$

A class with the minimum γ_{im} volume is assigned to course m .

This equation is applied for the courses with the next priority and remaining classrooms. The classroom with the minimum γ_{im} is selected. At this point, the question *Does the lecturer of this course have another course to attend in the same day in another classroom assigned beforehand?* If yes, then assign the course to the prior classroom, otherwise, assigned course to the classroom of concern.

This process must continue until all are classrooms assigned courses.

Note 5: Not more than one course should be assigned to a classroom at any time.

Note 6: The capacity of the classroom should be equal or greater than the course capacity.

The phases of the three-stage heuristic algorithm are flow-charted in Fig. 1:

In order to implement this algorithm the MATLAB s/w is applied.

4. Indexes for assessing the UCTP result

Here, the quality of this proposed method is assessed through the following seven indexes. Attempt is made here to appraise

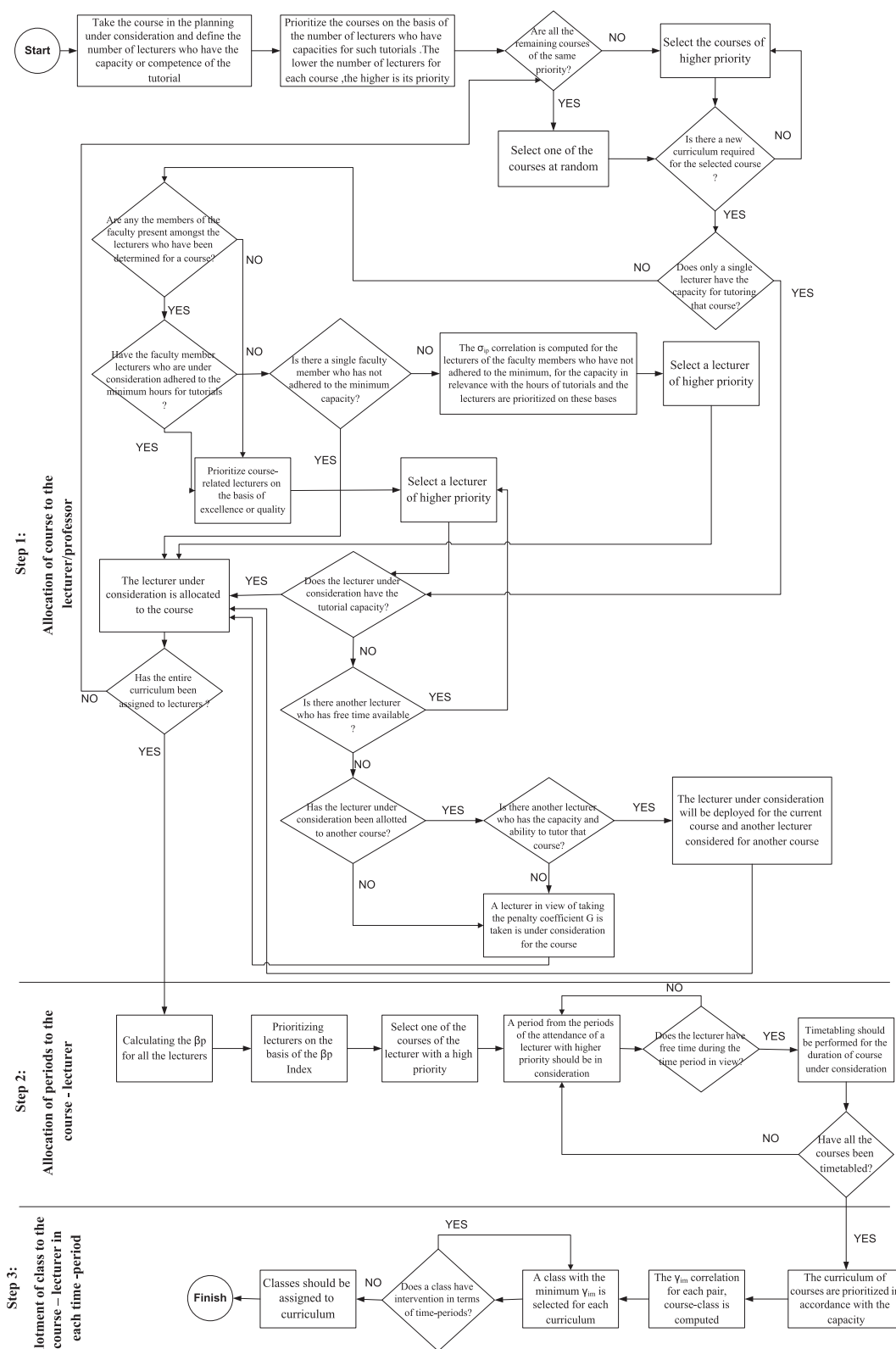


Fig. 1 Phases of the three-stage heuristic algorithm.

the qualitative dimensions in solving the UCTP, and evaluating the results. The advantages of these indexes are in assessing response of other methods, consist of mathematical models and meta-heuristic algorithms in solving the UCTP.

- Days of lecturers' attendance to total days of planning ratio

$$S_i = \sum_i \frac{\text{Simultaneous prerequisite and main courses groups for course } i}{\text{Total groups required for prerequisite and main courses for course } i}$$

Lecturers are specified as one of the most crucial resources of educational planning, thus, appropriate planning in this context can be contributive in their effectiveness. The compactness of their programs and an absence of intervals between their classes schedule can be effective on their efficiency, that is, the more their occupied hours, the more their attendance, hence a reduction in time and money spend as to, to and from university.

At this stage the following ratio is obtained through this index:

$$ND_i = \sum_p \frac{\text{Days planned for Lecturer/Professor } p}{\text{Total number of days planned}}$$

According to this definition, the lower this index, the better the planning.

- Lecturer quality level in planning context

As mentioned in the prior sections, one of the most crucial objective in educational planning is elevating its quality, in a sense that all assigned courses in any program be contributive in providing a higher level of excellence and quality, which will

$$Ca = \sum_{\text{Classroom}} \frac{\text{Classroom capacity} - \text{Course capacity assigned to a classroom} - \text{Classroom capacity}}{\text{Classroom capacity}}$$

obviously meet with the educational planning objectives. To accomplish this, this index is applied to assess the level of quality of the lecturers as to the planning; and in this manner, the quality ratio of the lecturer, in the planning for each course, is considered as the highest quality of the lecturer for that course.

$$Q_i = \sum_i \frac{\text{quality of the lecturer for the course } i}{\text{Highest quality for the lecturer for that course}}$$

According to this definition, the higher this index, the better the planning.

- Offering the simultaneous prerequisite and main courses to the all course ratio

In this index the simultaneous presentation of both, the prerequisite and main courses are of concern which is calculated through the following equation:

According to this definition, the higher this index, the better the planning.

Note 7: Make sure that, these courses cannot be taken by students simultaneously.

- Courses presented in the last time-slot to total courses ratio

In this index the courses offered in the last time-slot for each day is assessed, where the proportion of the courses given during the last time-slot is of concern as to the total offered courses.

$$R = \frac{\text{Courses offered in the last time - slot}}{\text{Total of courses offered}}$$

According to this definition, the smaller this index, the better the planning.

- Surplus classroom capacity to classroom capacity ratio

In this index, the surplus capacity of each classroom is assessed for each classroom and is calculated through the following equation:

According to this definition, the smaller this index, the better the planning.

- Full-time lecturer to total lecturers ratio

In this index the ratio of the employed full-time lecturer to total lecturers is calculated; and for this purpose in relevance to the number of courses of the full-time lecturers to that of the total number of courses offered is calculated through the following equation.

$$T = \frac{\text{courses assigned to Full - time lecturers}}{\text{Total courses}}$$

Table 2 Details of the postgraduate courses.

Course Code	Name of Course	Capacity	Number of Units	Number of Groups
1	Introduction to Industrial Engineering	35	2	1
2	Engineering Statistics	25	3	2
3	Econometrics	25	3	1
4	Engineering Economics	35	3	1
5	Strategic Planning	30	3	3
6	Advanced Linear Programming	35	3	1
7	Industrial Operations Planning	30	3	1
8	Time-series	30	3	1
9	Operational research	30	3	2
10	MCDM	30	3	2
11	Research Methodology	30	2	3
12	Statistical Method	35	3	2
13	Technical Terminology	30	2	2
14	Management and Leadership	30	3	3
15	Seminar	35	2	2
16	Dynamic System	30	3	1
17	Queue	25	3	2
18	Designing an automatic production system	25	3	1
19	Designing of Industrial Systems	30	3	3
20	Reliability	30	3	1
21	Fuzzy	25	3	1
22	Selection models of Investment Portfolios	25	3	1
23	Operation Management	30	3	2
24	Quality Management	30	3	2
25	Human Resources	30	3	3
26	Management & Risk Analysis	25	3	1
27	Selected Issues in Industries	30	3	3

Table 3 Details and the Lecturers attendance program.

Lecturer code	Minimizing unit of lecturer	maximizing unit of lecturer	Tuesday		Wednesday		Thursday	
			before noon	After noon	before noon	After noon	before noon	After noon
1	7	10	✓	✓	✓	✓	✓	✓
2	0	9			✓	✓		
3	0	9	✓	✓				
4	0	6						✓
5	0	12	✓	✓	✓	✓	✓	✓
6	3	9	✓	✓	✓	✓	✓	✓
7	0	6				✓		
8	10	16	✓	✓	✓	✓	✓	✓
9	10	16		✓		✓	✓	✓
10	10	16	✓	✓	✓	✓	✓	✓
11	0	9	✓	✓				
12	10	16	✓		✓		✓	
13	0	12	✓	✓				
14	6	10				✓	✓	✓
15	0	6						✓
16	0	6	✓				✓	
17	6	10					✓	✓
18	0	3					✓	✓
19	0	6			✓			
20	0	6					✓	
21	0	6		✓				
22	0	6					✓	

Table 4 Details of the Classroom details.

Class Code	1	2	3	4	5	6	7	8	9	10	11	12
Capacity	30	40	50	60	50	50	50	50	50	50	50	50

Table 5 The tutorial quality of lecturers.

Lecturer	Classroom																										
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27
1	0.93	0.78	0	0	0	0	0	0	0.82	0	0	0	0.87	0	0.8	0	0	0	0.84	0	0	0	0.85	0.82	0	0	0.91
2	0	0.87	0	0	0	0	0	0	0	0	0.86	0.8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.85
3	0	0	0.9	0.75	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	0.74	0	0	0.82	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.8
5	0	0	0	0	0.78	0	0	0	0	0	0	0	0	0.89	0	0	0	0	0	0	0	0	0	0	0.85	0	0
6	0.68	0	0	0	0.72	0	0.75	0	0.71	0	0	0	0	0.65	0	0	0	0	0	0	0	0	0.8	0	0.81	0	0
7	0	0	0	0	0.92	0	0	0	0	0	0	0	0	0.79	0	0	0	0	0	0	0	0	0	0	0.89	0	0
8	0.89	0	0	0.8	0	0.9	0	0	0	0.94	0.85	0	0	0	0.9	0	0	0	0.8	0.75	0	0	0.95	0.9	0	0	0.95
9	0.71	0	0	0	0.7	0	0.86	0	0	0	0.82	0	0	0	0.82	0	0	0	0	0	0	0	0	0	0	0	0.85
10	0.75	0.78	0	0	0.68	0	0	0	0.89	0	0	0	0.89	0	0	0.8	0	0	0	0	0.87	0	0	0	0	0	0.89
11	0.85	0.8	0	0	0	0	0	0	0	0.81	0	0	0	0	0	0	0.78	0	0	0	0	0	0	0	0	0	0.9
12	0.83	0	0	0	0	0	0	0	0.84	0	0	0.95	0	0	0	0.85	0.72	0	0	0	0	0	0	0	0	0	0.86
13	0.8	0	0	0	0	0	0	0	0	0	0	0	0.93	0	0	0	0	0.85	0.9	0	0	0	0	0	0	0	0.92
14	0	0.7	0	0.8	0	0	0	0	0	0	0	0.87	0	0	0	0	0	0	0.94	0.95	0	0	0	0	0	0	0.88
15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.85	0	0	0	0.78	0.78
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.87	0.79	0	0	0.82
17	0.86	0	0	0	0	0	0	0	0	0	0.89	0	0.84	0	0	0	0	0	0	0	0	0	0	0.93	0	0	0.9
18	0.71	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.86	0	0.72
19	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.84
20	0	0	0	0	0	0	0	0	0.75	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.92
21	0	0	0	0	0	0	0	0.9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.81
22	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.9	0	0	0	0	0	0	0.9

Table 6 The 5 solutions obtained through this three-stage heuristic algorithm.

Solution order	Ratio of days of attendance by Lecturers (ND _L) (–)	Level of excellence or quality of lecturers in planning (Q _L) (+)	Graduate and Postgraduate Course Ratio (S _L) (+)	Ratio of courses presented in the last period (R) (–)	Ratio of surplus class capacity (Ca) (–)	Ratio of lecturers (T) (+)	Ratio of the number of classrooms utilized by each Lecturer (Mo) (–)
Solution 1	7.51	44.61	6.042	0.25	15.291	0.721	3.25
Solution 2	7.33	45.95	5.919	0.125	13.267	0.667	3
Solution 3	8.5	44.97	5.27	0.125	17.273	0.526	3.5
Solution 4	7.96	45.27	5.094	0.2	13.457	0.417	3
Solution 5	7.28	45.74	5.827	0.075	14.539	0.627	3.6

According to this definition, the higher this index, the better the planning.

- Number of classrooms assigned to each lecturer to total available classroom ratio

The relocation of the lecturers among classrooms is assessed based on the number of classrooms assigned to each lecturer and is calculated through the following equation:

$$Mo = \sum_{\text{lecturer}} \frac{\text{classrooms assigned to each lecturer}}{\text{total available classrooms}}$$

According to this definition, the smaller this index, the better the planning.

5. The graduate studies courses timetabling for faculty of Industrial engineering Department of the Islamic Azad University of Najafabad

According to the available data academic algorithms and methods presented by researchers can only be of acceptance when, they are feasible and capable of having implementation abilities. For this purpose, in this phase, a three-stage algorithm is introduced for solving UCTP, with a focus on post graduate study courses in Islamic Azad University, Industrial Engineering Department, Najafabad branch, where the M.sc students and Ph.D candidate are involved. The details of the related courses in this program are tabulated in Table 2.

The detail of lecturer attendance are tabulated in Table 3.

The course itemize in Table 2 must be taught in 12 assigned classroom subject to the schedule presented in Table 4.

This proposed algorithm, which is a qualitative approach, includes the tutorial quality of the lecturers capable of teaching the given course, according to the student feedback at the end of each term, expressed in Table 5.

In this case study a three-stage algorithm is fed to MATLAB s/w to be coded through which the UCTP of concern is solved. In this algorithm all selection processes are run in a random manner, consequently, where the responses are obtained after 50 iteration, which are then compared pairwise solutions manner.

After this process, the 5 best responses are selected, to be assessed through the 7 indexes, the details of which are tabulated in Table 6.

To evaluate the assessed 5 solutions with respect to the 7 indexes, they are compared pairwise solution manner. In this pairs, if one solution is better than more indexes, it is chosen and the other is eliminated; this process runs until the best solution is reviled. In accordance with this method, Solution 2 is proved to have more appropriate response to UCTP of the subject university. Subsequently, the postgraduate course timetabling of the Industrial Engineering Department is extracted according to Solution 2, Table 7.

As observed in Table 7, 48 curriculums must be included in the timetabling of postgraduate courses of the industrial engineering Department, which is scheduled. The days of lecturers' attendance, consisting of 22 lecturer, to total days of planning ratio is 7.33 in the planning. This finding indicates that each lecturer has approximately one day of teaching in the course timetabling. As to lecturer quality in offering the course to the lecturer quality level in planning context quality in teaching the courses is 45.95, that is quality of each course is 96% with respect to optimum conditions. The index of courses presented in the last time-slot to total courses ratio is 0.125, which is an appropriate condition. The index of surplus classroom capacity to classroom capacity ratio is 13.267, indicating that in each classroom, there is a surplus capacity for 13 student on an average, which is an appropriate figure. The index of full-time lecturer to total lecturers' ratio is 0.667, indicating that one-thirds of the lecturers are part-time lecturer, which in turn indicates that in this course timetabling planning internal resources are applied. The index of number of classrooms assigned to each lecturer to total available classroom ratio is 3, which is an appropriate level.

6. Discussion and conclusion

The issue of planning and timetabling the daily affairs consist of factors, which can have an impact on the quality of life for individuals. It is obvious that, one of the crucial day to day matters of a social population is related to higher education in universities. The UCTP of the courses can have a direct impact on the educational quality of a given society. In the

Table 7 Postgraduate course timetabling of the Industrial Engineering Department.

Time slot day	Time Slot 1	Time Slot 2	Time Slot 3	Time Slot 4	Time Slot 5	Time Slot 6	Time Slot 7	Time Slot 8	Time Slot 9	Time Slot 10	Time Slot 11	Time Slot 12
Tuesday	Course 14 (lecture 5)(classroom 207)			Course 10 (lecture 8)(classroom 207)			Course 5 (lecture 5)(classroom 207)			Course 5 (lecture 5)(classroom 207)		
	Course 6 (lecture 8)(classroom 208)			Course 14 (lecture 5)(classroom 208)			Course 13 (lecture 10)(classroom 208)		Course 1 (lecture 1)(classroom 208)			
	Course 15 (lecture 1)(classroom 209)			Course 16 (lecture 10)(classroom 209)			Course 7 (lecture 9)(classroom 209)			Course 4 (lecture 8)(classroom 209)		
	Course 14 (lecture 6)(classroom 211)			Course 19 (lecture 13)(classroom 211)			Course 8 (lecture 21)(classroom 211)			Course 15 (lecture 9)(classroom 211)		
			Course 21 (lecture 1)(classroom 212)				Course 10 (lecture 8)(classroom 212)			Course 27 (lecture 13)(classroom 212)		
	Course 3 (lecture 3)(classroom 213)					Course 23 (lecture 1)(classroom 213)			Course 13 (lecture 10)(classroom 213)			
	Course 17 (lecture 12)(classroom 214)			Course 17 (lecture 12)(classroom 214)			Course 19 (lecture 13)(classroom 214)					
	Course 18 (lecture 13)(classroom 215)											
	Course 21 (lecture 10)(classroom 216)											
Wednesday	Course 9 (lecture 10)(classroom 207)			Course 9 (lecture 12)(classroom 207)			Course 11 (lecture 9)(classroom 207)		Course 11 (lecture 9)(classroom 207)			
	Course 12 (lecture 12)(classroom 208)			Course 2 (lecture 2)(classroom 208)			Course 20 (lecture 14)(classroom 208)			Course 12 (lecture 14)(classroom 208)		
	Course 28 (lecture 8)(classroom 209)			Course 19 (lecture 14)(classroom 209)			Course 25 (lecture 7)(classroom 209)			Course 25 (lecture 7)(classroom 209)		
	Course 2 (lecture 2)(classroom 211)											
Thursday	Course 5 (lecture 9)(classroom 207)					Course 24 (lecture 17)(classroom 207)				Course 26 (lecture 15)(classroom 207)		
	Course 11 (lecture 17)(classroom 208)		Course 24 (lecture 17)(classroom 208)				Course 22 (lecture 15)(classroom 208)					
	Course 25 (lecture 18)(classroom 211)											
	Course 27 (lecture 20)(classroom 212)											

UCTP, there are usually a number of educational resources consisting of lecturers, courses, classrooms and students assigned to a series of time slots and in this assignment, a number of hard and soft constraints are of concern.

In the UCTP, the hard constraints are absolutely satisfied; and the soft constraints are also met with at utmost extent. With respect to the discussion on this context, a three-stage heuristic algorithm is proposed to solve UCTP. Consequently, the course assignment to the lecturer is takes place in the first stage, the courses assigned to the lecturers are scheduled in the second stage and in the third stage, a classroom is assigned to each course. To assess the implementation capabilities of this three-stage heuristic algorithm, a UCTP is solved by applying the Postgraduate of the Industrial engineering department of Islamic Azad University, Najafabad branch as a case study. The findings here are evaluated through the 7 solution validation indexes.

In accordance with the objective of this study, as to solving this UCTP with a qualitative-oriented approach, the 7 solution validation indexes are determined to evaluate a quality of planning. The index of the days of lecturers' attendance to total days of planning ratio is 7.33 in the planning. This finding indicates that each lecturer has approximately one day of teaching in the course timetabling. The index of the lecturer quality in offering the course to the lecturer quality level in planning context quality in teaching the courses is 96% with respect to optimum conditions. The index of courses presented in the last time-slot to total courses ratio is 0.125, which is an appropriate condition. The index of surplus classroom capacity to classroom capacity ratio indicates that a surplus capacity of 13 student on an average, an appropriate figure. The index of full-time lecturer to total lecturers' ratio is 0.667, indicating that

one-thirds of the lecturers are part-time lecturer, which in turn indicates that in this course timetabling planning internal resources are applied. The index of number of classrooms assigned to each lecturer to total available classroom ratio is 3, which is an appropriate level.

There exist many studies with the focus on UCTP by Akkan and Gulcu [16], Bagger et al. [15], Gulcu and Akkan [17] and Myers and Yellen [18], to the findings of whom the finding here are compared with. In their studies they have applied approximate method to solve problems of big dimensions with presenting entire curriculums of an individual course, planning standard curriculums based on the hours of attendance, refrain from intermingling in the program of lecturers, Refraining from intermingling with respect to the location of classrooms and running classes through the week in a compact manner as constraints. In this study with respect to the real scale of NP-Hard problem the approximate method are applied as well by considering the same constraint in addition to presenting some new constraint consisting of Prioritization of full-time lecturer to part-time, Lecturers' quality of teaching and Utilizing the lecturer presence time at its maximum, which lead to a higher quality of timetabling with better performance vs. its counterparts, thus, a promotion in educational quality.

The results of this study indicate that by applying this algorithm, the educational planning will be of better efficiency and compactness; because the lecturers and students would have a less number of days to attend university. In this approach the direct and indirect transportation cost is reduced for both the parties. By the compactness in lecturer schedule planning, consultation prospects enhance between the lecturers and the students and in this mode the educational quality of the university

can be increased. In this algorithm about 70% of the full-time lecturers are applied in timetabling and university costs are reduced and internal capacities are utilized to a greater extent. The result obtained here indicate that appropriate responses could be yield through this algorithm.

At this point, it should be mention that despite the advantages of proposed algorithm there exist some limitations like, here no differentiation is concerned between mandatory and non-mandatory courses and the fact that course presentation is uniform in different daily time slots, while in order to promote educational quality, have less courses at the end of daily time slots.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

- [1] E.K. Burke, B. McCollum, A. Meisels, S. Petrovic, R. Qu, A graph-based hyper-heuristic for educational timetabling problems, *Eur. J. Oper. Res.* 176 (1) (2007) 177–192.
- [2] D. Zhang, Y. Liu, R. M'Hallah, S.C.H. Leung, A simulated annealing with a new neighborhood structure based algorithm for high school timetabling problems, *Eur. J. Oper. Res.* 203 (3) (2010) 550–558.
- [3] S. Petrovic, E.K. Burke, University timetabling, In: J. Leung (Ed.), *Handbook of Scheduling: Algorithms, Models, and Performance Analysis*, CRC Press, 2004. (Chapter 45).
- [4] A. Elhag, E. Ozcan, A grouping hyper-heuristic framework: application on graph colouring, *Expert Syst. Appl.* 42 (13) (2015) 5491–5507.
- [5] H. Babaei, J. Karimpour, A. Hadidi, A survey of approaches for university course timetabling problem, *Comput. Ind. Eng.* 86 (1) (2014) 43–59.
- [6] M.R. Feizi-Derakhshi, H. Babaei, J. Heidarzadeh, A survey of approaches for university course timetabling problem, in: *Proceedings of 8th International Symposium on Intelligent and Manufacturing Systems (IMS 2012)*, Sakarya University Department of Industrial Engineering, Adrasan, Antalya, Turkey, 2012, pp. 307–321.
- [7] J.H. Obit. Developing Novel Meta-heuristic, Hyper-heuristic and Cooperative Search for Course Timetabling Problems, Ph. D. Thesis, School of Computer Science University of Nottingham, 2010.
- [8] V. Cacchiani, A. Caprara, R. Roberti, P. Toth, A new lower bound for curriculum-based course timetabling, *Comput. Oper. Res.* 40 (10) (2013) 2466–2477.
- [9] R. Lewis, J. Thompson, Analysing the effects of solution space connectivity with an effective metaheuristic for the course timetabling problem, *Eur. J. Oper. Res.* 240 (3) (2015) 637–648.
- [10] J.A. Soria-Alcaraz, E. Özcan, J. Swan, G. Kendall, M. Carpio, Iterated local search using an add and delete hyper-heuristic for university course timetabling, *Appl. Soft Comput.* 40 (1) (2016) 581–593.
- [11] I. Méndez-Díaz, P. Zabala, J.J. Miranda-Bront, An ILP based heuristic for a generalization of the post-enrollment course timetabling problem, *Comput. Oper. Res.* 76 (1) (2016) 195–207.
- [12] R. Borchani, A. Elloumi, M. Masmoudi, Variable neighborhood descent search based algorithms for course timetabling problem: application to a Tunisian University, *Electron. Notes Discrete Math.* 58 (1) (2017) 119–126.
- [13] V.I. Skoullis, I.X. Tassopoulos, G.N. Beligiannis, Solving the high school timetabling problem using a hybrid cat swarm optimization based algorithm, *Appl. Soft Comput.* 52 (1) (2017) 277–289.
- [14] K. Song, S. Kim, M. Park, H.S. Lee, Energy efficiency-based course timetabling for university buildings, *Energy* 139 (1) (2017) 394–405.
- [15] N.C.F. Bagger, M. Sorensen, T.R. Stidsen, Benders' decomposition for curriculum-based course timetabling, *Comput. Ind. Eng.* 91 (1) (2018) 178–189.
- [16] C. Akkan, A. Gulcu, A bi-criteria hybrid Genetic Algorithm with robustness objective for the course timetabling problem, *Comput. Oper. Res.* 90 (1) (2018) 22–32.
- [17] A. Gulcu, C. Akkan, Bi-criteria simulated annealing algorithms for the robust university course timetabling problem, in: *Proceedings of the 12th International Conference on the Practice and Theory of Automated Timetabling*, Vienna, Austria, August 2831, 2018, pp. 129–136.
- [18] D.S. Myers, J. Yellen, A multi-objective timetabling system that facilitates scheduling across academic programs, in: *Proceedings of the 12th International Conference on the Practice and Theory of Automated Timetabling (PATAT-2018)*, Vienna, Austria, August 2831, 2018, pp. 153–170.
- [19] E.I. Ásgeirsson, P. Gunnarsdóttir, An integer programming formulation for the music school timetabling problem, in: *Proceedings of the 12th International Conference on the Practice and Theory of Automated Timetabling (PATAT-2018)*, Vienna, Austria, August 2831, 2018, pp. 171–182.
- [20] M.S. Ramalho, Constraint definition in student scheduling with adversarial behaviour in mind; A case study of a multi objective optimization problem in the University of Porto, in: *Proceedings of the 12th International Conference on the Practice and Theory of Automated Timetabling (PATAT-2018)*, Vienna, Austria, August 2831, 2018, pp. 199–214.
- [21] S.M. Al-Yakoob, H.D. Sherali, Mathematical Models and Algorithms for a High school timetabling problem, *Comput. Oper. Res.* 61 (1) (2015) 56–68.
- [22] H. Vermuyten, S. Lemmens, I. Marques, J. Belien, Developing compact course timetables with optimized student flows, *Eur. J. Oper. Res.* 251 (2) (2016) 651–661.
- [23] R. Bellio, S. Ceschia, L.D. Gaspero, A. Schaerf, T. Urli, Feature-based tuning of simulated annealing applied to the curriculum-based course timetabling problem, *Comput. Oper. Res.* 65 (1) (2016) 83–92.
- [24] G.H.G. Fonseca, H.G. Santos, E.G. Carrano, Integrating matheuristics and metaheuristics for timetabling, *Comput. Ind. Eng.* 74 (1) (2016) 108–117.
- [25] L. Saviniec, A.A. Constantino, Effective local search algorithms for high school timetabling problems, *Appl. Soft Comput.* 60 (1) (2017) 363–373.

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